TRIBHUVAN UNIVERSITY

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THAPATHALI CAMPUS

A Project Proposal

On

Gamified Learning Assessment System

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ABSTRACT

The Gamified Learning Assessment System (GLAS) is an innovative web-based platform designed to transform traditional academic evaluation into an engaging, motivational learning experience through game mechanics. Traditional assessment methods often fail to engage digital-native students, creating anxiety and focusing on grades rather than learning. GLAS addresses these challenges by implementing game design principles; experience points (XP), levels, achievements and quests within an educational context.

This full-stack web application utilizes PostgreSQL for robust data management, Node.js with Express.js for backend API development, and React.js for an interactive frontend interface. The system features role-based access for students, teachers, and administrators, each with customized dashboards. Students engage with learning materials through ”quests,” earn XP for completed assignments, unlock achievements, and progress through levels.

Key technical innovations include a dynamic XP calculation engine. The system maintains academic integrity through secure authentication using JWT tokens, bcrypt password hashing, and role-based permissions while providing educators with detailed analytics on student engagement and performance.

GLAS demonstrates how thoughtfully applied gamification can increase student motivation, provide meaningful feedback, and create a positive learning culture focused on growth and mastery.

This project showcases the successful implementation of modern web technologies to address real educational challenges, providing a scalable, secure, and engaging platform that bridges the gap between traditional assessment and contemporary student engagement strategies.

Keywords: Gamification, Learning Assessment, Experience Points, PostgreSQL, React, Node.js, Educational Technology, Student Engagement

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List of Abbreviations

API Application Programming Interface.

CRUD Create, Read, Update, Delete.

ERD Entity Relationship Diagram.

GLAS Gamified Learning Assessment System.

GUI Graphical User Interface.

HTTP Hypertext Transfer Protocol.

JWT JSON Web Token.

MVC Model-View-Controller.

REST Representational State Transfer.

SQL Structured Query Language.

XP Experience Points.

1. INTRODUCTION

1.1 Background

Traditional learning assessment systems are predominantly characterized by rigid, high-stakes examinations and assignment submissions that often induce student anxiety and prioritize rote memorization over genuine understanding. In an era dominated by digital interactivity and gamified experiences in everyday applications, these conventional methods fail to leverage the motivational power of game mechanics. Gamification, the application of game-design elements in non-game contexts, has shown significant potential in enhancing engagement, motivation, and productivity in fields like marketing, fitness, and corporate training. The education sector stands to benefit immensely from this approach by transforming assessment from a source of stress into a compelling and rewarding journey of mastery.

1.2 Motivation

The primary motivation for developing the Gamified Learning Assessment System (GLAS) stems from the observable disengagement and anxiety associated with traditional grading systems. Current Learning Management Systems (LMS) often function as digital file repositories and gradebooks, lacking intrinsic motivational elements. GLAS is driven by the vision of creating a learning environment where assessment is continuous, feedback is immediate and constructive, and progress is visually rewarded. By integrating proven game mechanics such as Experience Points (XP), levels, badges, the system aims to foster a growth mindset, increase time-on-task, and make the learning process more transparent and enjoyable for the digital-native generation.

1.3 Problem Defination

The core problem addressed by GLAS is the lack of student engagement and intrinsic motivation in traditional learning assessment frameworks. Specific issues include:

1. Passive Learning Experience: Students are passive recipients of grades rather than active participants in a learning journey.
2. Anxiety and Fear of Failure: High-stakes exams create stress, discouraging risk-taking and exploration.
3. Lack of Immediate Feedback: Delayed grading cycles prevent students from promptly correcting misunderstandings.
4. Focus on Grades over Mastery: The system incentivizes achieving a final score rather than understanding concepts and demonstrating skill progression.
5. One-Size-Fits-All Approach: Traditional assessments rarely adapt to individual

learner paces or provide personalized challenge pathways.

1.4 Objectives

The development of GLAS is guided by the following primary and secondary objectives:

1.4.1 Primary Objective

To design and implement a fully functional, web-based gamified learning assessment system that increases student engagement and provides a more holistic view of learner progress through game mechanics.

1.4.2 Secondary Objectives

1. To develop a secure, role-based platform with distinct interfaces for Students, Teachers, and Administrators.
2. To implement core gamification features: an XP/Leveling system, a badge/achievement system, quest-based learning modules, and dynamic leaderboards.
3. To create robust backend services (APIs) for user management, content delivery, progress tracking, and report generation using Node.js, Express, and PostgreSQL.
4. To build an intuitive, responsive frontend application using React.js that provides a seamless and engaging user experience.
5. To ensure system security through JWT-based authentication, bcrypt password hashing, and SQL injection prevention.

1.5 Scope and Applications

1.5.1 Inclusions

The GLAS project scope includes:

- A multi-role web application (Student, Teacher).
- User authentication and authorization system.
- Student Dashboard: Displays active quests, progress map, XP/Level, achievements.
- Teacher Dashboard: Enables creation/managing of courses, quests (assignments), and review of student submissions with XP awarding.
- Core Gamification Engine: Automated XP calculation, level progression, and badge awarding logic.
- Generation of progress reports in PDF format.
- A relational database (PostgreSQL) to store all system data.

- RESTful API backend serving the frontend React application.

1.5.2 Exclusions

The following are explicitly out of scope for this version:

- Development of native mobile applications (iOS/Android).
- Integration with external Learning Management Systems (LMS) like Moodle or Canvas.
- Advanced AI-driven features for personalized quest generation or automated essay grading.
- In-built communication tools like forums or direct messaging.
- Offline functionality or mobile responsiveness beyond standard web practices.
- Payment gateways or monetization features.

2. METHODOLOGY

2.1 System Development Life Cycle

The GLAS project will follow an Agile-inspired iterative methodology with elements of the Prototype model. The core phases—Requirement Analysis, Design, Implementation, Testing, and Deployment—will be revisited iteratively throughout the project lifecycle.

2.2 Technology Stack Selection

The technology stack has been selected based on performance, scalability, developer ecosystem, and project requirements.

- Backend Runtime & Framework: Node.js with Express.js. Node.js offers a non-blocking, event-driven architecture ideal for real-time features like leaderboards. Express.js provides a minimal and flexible framework for building robust RESTful APIs.
- Database: PostgreSQL. A powerful, open-source relational database chosen for its reliability, ACID compliance, and advanced features like JSON support, which is suitable for storing dynamic game state or achievement criteria.
- Database Driver: pg (node-postgres). A low-level, performant PostgreSQL client for Node.js. We will use raw SQL queries instead of an ORM for finer control over database interactions, optimized performance, and a deeper understanding of the data layer.
- Frontend Library & Framework: React.js. A component-based library for building dynamic and responsive user interfaces. Its virtual DOM ensures efficient updates, crucial for a data-rich, interactive dashboard.
- HTTP Client: Axios. A promise-based HTTP client for making API calls from the React frontend to the Node.js backend. It simplifies handling asynchronous requests and responses.
- Styling: CSS with Tailwind CSS. A utility-first CSS framework that enables rapid UI development and ensures a consistent, responsive design without writing custom CSS for every element.
- Authentication: JSON Web Tokens (JWT) for stateless session management and bcrypt for secure password hashing.
- PDF Generation: PDFKit. A Node.js library for generating complex, data-driven PDF documents such as student progress reports and transcripts.
- Development Tools: Nvim as the primary code editor, Git for version control, GitHub for repository hosting and collaboration.

2.3 System Architecture

GLAS will follow a client-server architecture with a clear separation between the frontend (Presentation Layer), backend (Application Layer), and database (Data Layer).

- Client Layer (React Frontend): Serves the interactive UI. Communicates with the backend via HTTP requests (Axios) to perform all operations.
- Server Layer (Node.js/Express API): Contains the business logic, handles authentication, processes requests, executes database queries, and implements the gamification engine (XP calculation, badge checks).
- Data Layer (PostgreSQL): Persistently stores all structured data including user profiles, courses, quests, submissions, XP transactions, and achievement records.
- The architecture will be RESTful, ensuring stateless communication and scalability.

2.4 Database Design

The database schema will be designed using a normalized relational model to ensure data integrity and avoid redundancy. Key entities include:

- users: Stores login credentials (hashed), role, profile info.
- courses: Contains course details, linked to a teacher.
- enrollments: Manages the many-to-many relationship between students and courses.
- quests: Represents assignments/tasks, with details like title, description, max XP, deadline, and course_id.
- submissions: Records student submissions for quests, with status (pending/graded) and teacher feedback.
- xp_transactions: An audit trail for all XP earned or spent, linked to user and quest/submission.
- user_levels: Tracks a user's current level and total cumulative XP.
- achievements & user_achievements: Defines available badges and tracks which users have unlocked them.

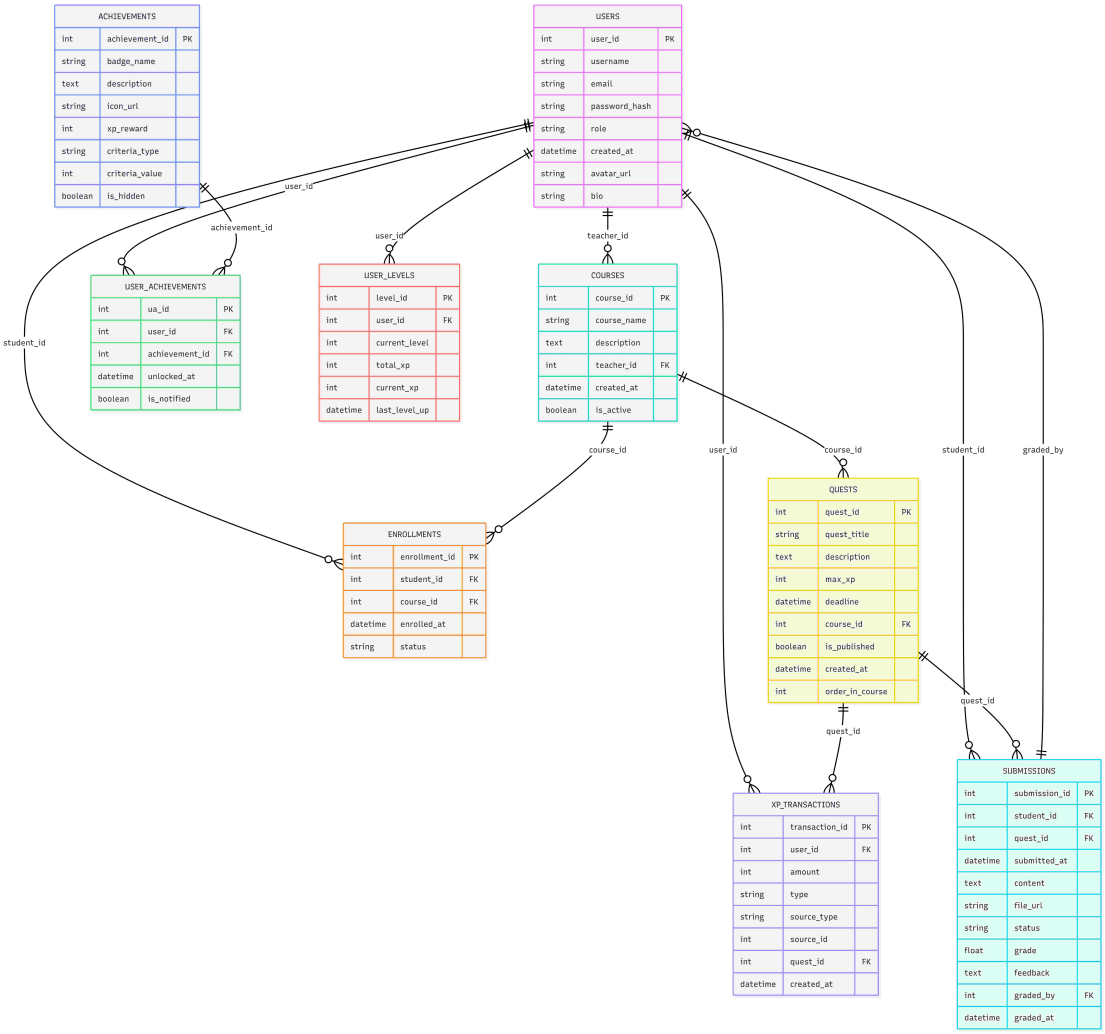

Figure 2-1: ER diagram of database

3. EXPECTED OUTCOME

3.1 Functional Features

Upon completion, GLAS will be a deployable web application with the following key features:

- Secure, role-based login for Students, Teachers, and Admins.
- Teacher: Create/manage courses and ”quests” (assignments with XP values), grade submissions, and award XP.
- Student: Browse and enroll in courses, view and submit quests, track XP/Level progress, unlock and display achievements.
- Automated Gamification: Background processes that calculate level progression and trigger achievement awards based on defined rules (e.g., ”Complete 10 Quests,” ”Score 1000 total XP”).
- PDF report generation for student progress and course performance.

3.2 User Interfaces

- Student Dashboard: A visually engaging hub featuring a progress bar/level indicator, a list of ”Active Quests,” a collection of earned badges.
- Teacher Dashboard: A clean, functional interface with tabs for managing ”My Courses,” ”Create New Quest,” and ”Grade Submissions.”.

3.2.1 Student Dashboard

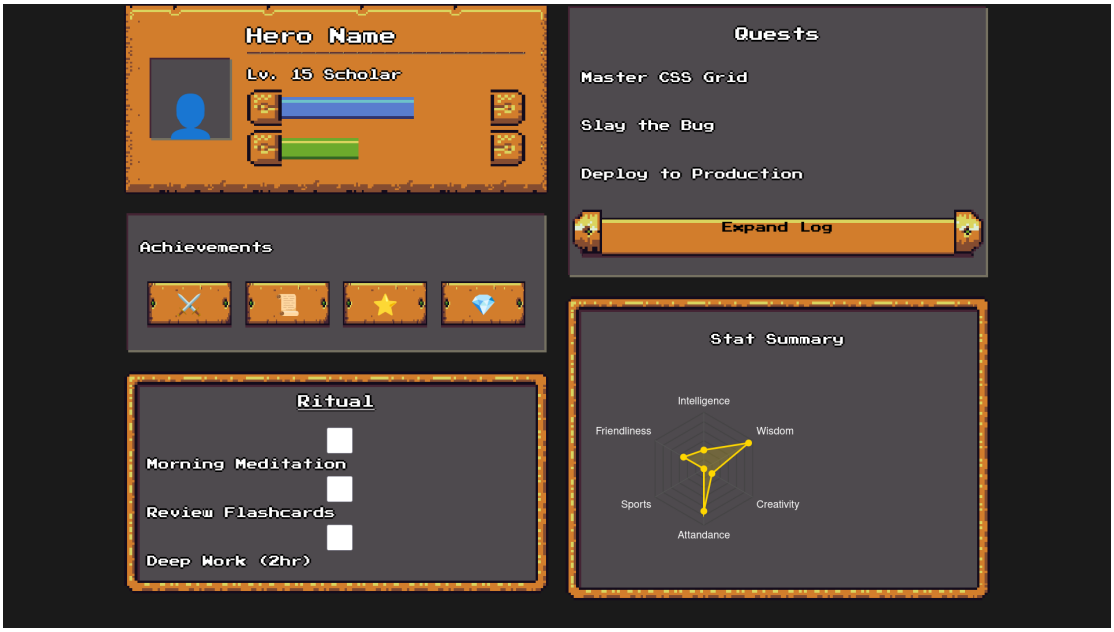

Figure 3-1: Student Dashboard

3.2.2 Teacher Dashboard

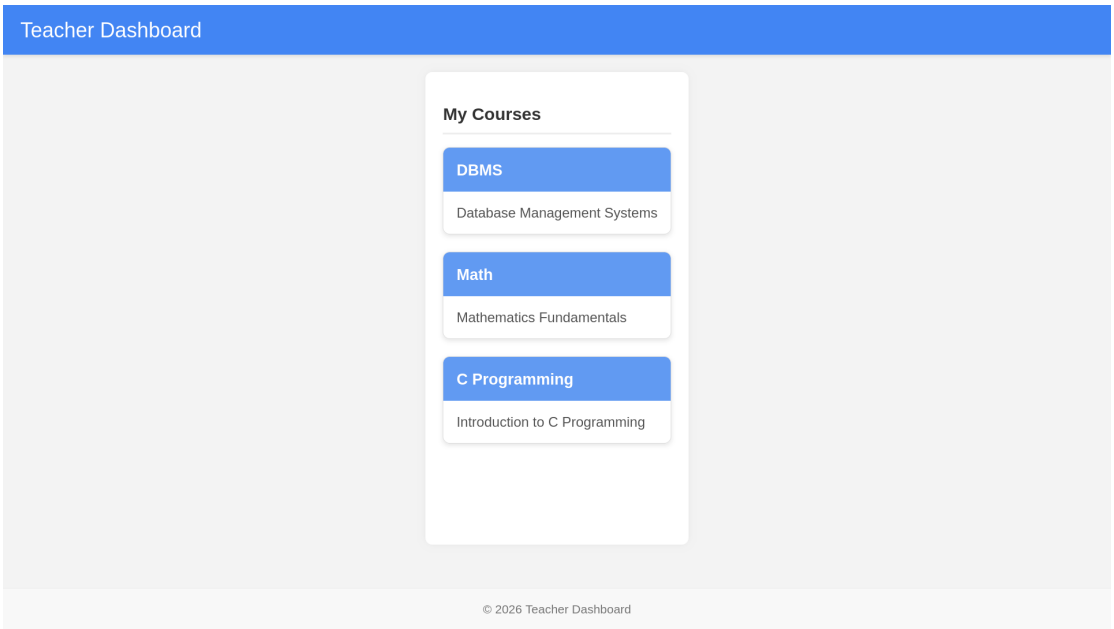

Figure 3-2: Teacher Dashboard

4. TIME ESTIMATION

Table 4-1: Project Timeline Gantt Chart (Starting January 1, 2026)

Jan 1-7	Jan 8-14	Jan 15-21	Jan 22-28	Jan 29-Feb 4

Color	Task
	Project Setup & Database Schema
	User Authentication & Authorization
	Core Course & Quest Management
	Gamification Engine
	Integration & Testing
	Documentation (ongoing)